Problem Set: Problem 2(ii)

Converting $e^-\mu^- \to e^-\mu^-$ to $q\,q' \to q\,q'$ via Gluon Exchange

PHYS 7502 — Particles and Nuclei 2

Question Statement and Deliverables

Problem 2(ii) — Verbatim

Also convert the Mandelstam variable amplitude to the equivalent one for $qq \to qq$ parton–parton (parton–parton x.s., ignoring/not including pdf and fragmentation effects) scattering via gluon exchange—presume different quark flavors to avoid the u-channel inclusion. In this case you may either sum/average, or choose specific color choices for the incoming and outgoing quarks and exchanged gluon.

What This Problem Wants

Starting from the QED result for $e^-\mu^- \to e^-\mu^-$ in Mandelstam variables, produce the QCD analog for $q\,q' \to q\,q'$ (different flavors, t-channel gluon exchange only). Specifically:

- Identify the Feynman rule substitutions: QED vertex → QCD vertex, photon propagator → gluon propagator.
- Compute the **color factor**, either by:
  (a) Averaging over initial colors and summing over final colors and gluon index ("sum/average"), *or*
  (b) Choosing specific color assignments for external quarks and the exchanged gluon.
- Write down $\langle|\mathcal{M}|^2\rangle$ for $q\,q' \to q\,q'$ in Mandelstam variables.

1 Starting Point: QED Result from Part (a)

Recap: $e^-\mu^- \to e^-\mu^-$ in Mandelstam Variables

In part (a) / Griffiths 7.38, we derived the spin-averaged, spin-summed amplitude squared for electron–muon scattering in the massless ($m_e, m_\mu \to 0$) limit. There is only one diagram (the t-channel photon exchange), and the result in Mandelstam variables is:

$$\boxed{\langle|\mathcal{M}|^2\rangle_{\text{QED}} = 2g_e^4\,\frac{s^2+u^2}{t^2}} \tag{1}$$

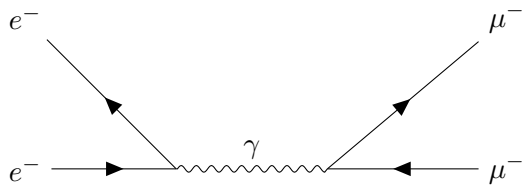
where $g_e^2 = 4\pi\alpha$, and s, t, u are the usual Mandelstam variables satisfying $s+t+u=0$ (massless limit).

Why Only a t-channel?

For $e^-\mu^- \to e^-\mu^-$, electron number and muon number are each conserved at every vertex. The only way to connect the electron line to the muon line is through a t-channel photon. There is no s-channel (that would require $e^-\mu^-$ annihilating into a photon, which violates lepton flavor) and no u-channel (the electron and muon are distinguishable).
The **exact same logic** applies to $q\,q' \to q\,q'$ with different flavors: a u-quark cannot turn into a d-quark at a QCD vertex (gluons are flavor-blind), so only the t-channel gluon exchange diagram exists.

2 Step 1: Identify the QED → QCD Substitutions

The $e^-\mu^- \to e^-\mu^-$ amplitude comes from the Feynman diagram:



Now we translate each piece to QCD:

QED vs. QCD Feynman Rules

Element	QED	QCD
Vertex factor	$ig_e\gamma^\mu$	$i\frac{g_s}{2}\,\lambda^a_{ji}\,\gamma^\mu$
Propagator	$\frac{-ig_{\mu\nu}}{q^2}$	$\frac{-ig_{\mu\nu}\,\delta^{ab}}{q^2}$
Coupling constant	$g_e = \sqrt{4\pi\alpha}$	$g_s = \sqrt{4\pi\alpha_s}$
Color structure	None	λ^a (Gell-Mann matrices)

Convention note: We use Griffiths' QCD vertex factor $\frac{g_s}{2}\lambda^a\gamma^\mu$. Thomson uses $\frac{g_s}{2}\lambda^a\gamma^\mu$ as well. Some texts absorb the $\frac{1}{2}$ into the definition of the generators $T^a = \frac{\lambda^a}{2}$ and write the vertex as $g_s T^a\gamma^\mu$.

Why the Gell-Mann Matrices?

In QED, the photon couples to electric charge, which is just a number ($\pm e$). In QCD, the gluon couples to *color charge*, and color has three components (r, g, b). The Gell-Mann matrices λ^a ($a = 1, \ldots, 8$) are the generators of $\mathrm{SU}(3)_{\text{color}}$ — they are 3×3 matrices that tell you how a quark's color rotates when it emits/absorbs a gluon of color index a. They play the same role as the Pauli matrices in SU(2).
Each vertex carries a color factor λ^a_{ji}, where i is the color of the incoming quark and j is the color of the outgoing quark. The gluon propagator δ^{ab} ensures that the same gluon color index appears at both vertices.

3 Step 2: Write the QCD Amplitude

The QCD amplitude for $q_i\, q'_k \to q_j\, q'_l$ via t-channel gluon exchange is obtained by direct substitution into the QED amplitude. Following Griffiths eq. (9.31):

$$\mathcal{M} = -\frac{g_s^2}{4}\frac{1}{q^2}\left[\bar{u}(p_3)\gamma^\mu u(p_1)\right]\left[\bar{u}(p_4)\gamma_\mu u(p_2)\right] \underbrace{\left(\frac{\lambda^a_{ji}\,\lambda^a_{lk}}{1}\right)}_{\text{color factor at amplitude level}} \tag{2}$$

where summation over the gluon index $a = 1, \ldots, 8$ is implied (from contracting δ^{ab} with the two vertex factors).

Where Does Everything Come From?

- **Two vertex factors:** Each contributes $i\frac{g_s}{2}\lambda^a\gamma^\mu$. Together: $\left(\frac{g_s}{2}\right)^2 = \frac{g_s^2}{4}$ and $\lambda^a_{ji}\lambda^a_{lk}$.
- **Gluon propagator:** $\frac{-ig_{\mu\nu}\delta^{ab}}{q^2}$, with $q = p_1 - p_3$ (t-channel momentum transfer, so $q^2 = t$).
- **Spinor structure:** Identical to $e^-\mu^- \to e^-\mu^-$: $[\bar{u}(3)\gamma^\mu u(1)][\bar{u}(4)\gamma_\mu u(2)]$.
- The δ^{ab} contracts the two λ^a factors, giving a single sum $\sum_a \lambda^a_{ji}\lambda^a_{lk}$.

Comparing with the QED amplitude:

$$\mathcal{M}_{\text{QED}} = -\frac{g_e^2}{q^2}\left[\bar{u}(3)\gamma^\mu u(1)\right]\left[\bar{u}(4)\gamma_\mu u(2)\right] \tag{3}$$

we see the **spinor/Dirac structure is identical**. The only differences are:

1. $g_e^2 \to \frac{g_s^2}{4}$ (coupling constant change, including the $\frac{1}{2}$ from each vertex).
2. An additional multiplicative color factor $\lambda^a_{ji}\lambda^a_{lk}$.

4 Step 3: Square the Amplitude and Handle Color

When we square $|\mathcal{M}|^2$ and perform spin averaging (Casimir's trick, traces, etc.), the spinor traces give us exactly the same Mandelstam-variable expression as in QED. The color part **factorizes**

completely from the Dirac algebra because the λ^a_{ji} are just numbers (matrix elements), not Dirac matrices.

$$|\mathcal{M}|^2 = \underbrace{\frac{g_s^4}{16}}_{\text{from } (g_s/2)^4} \times \underbrace{|\lambda^a_{ji}\lambda^a_{lk}|^2}_{\text{color}} \times \underbrace{\frac{1}{t^2}\,(\text{spinor traces})}_{\text{same as QED}} \tag{4}$$

Key Insight: Color Factorizes

The color matrices λ^a live in a completely different vector space (3-dimensional color space) from the Dirac spinors (4-dimensional spinor space). They don't interfere with each other algebraically, so we can compute the spin part and the color part independently and just multiply at the end.
This is exactly analogous to how in atomic physics, spin and orbital angular momentum calculations factorize when there's no spin-orbit coupling.

4.1 Option A: Sum/Average over Colors

We average over the initial quark colors ($3 \times 3 = 9$ combinations) and sum over the final quark colors and gluon index:

$$\langle|C|^2\rangle = \frac{1}{9}\sum_{i,j,k,l=1}^{3}\sum_{a,b=1}^{8}\frac{1}{16}\,\lambda^a_{ji}\lambda^a_{lk}\,(\lambda^b_{ji})^*(\lambda^b_{lk})^* \tag{5}$$

Since the Gell-Mann matrices are Hermitian ($\lambda^{a\dagger} = \lambda^a$), we have $(\lambda^b_{ji})^* = \lambda^b_{ij}$, so:

$$\langle|C|^2\rangle = \frac{1}{9}\cdot\frac{1}{16}\sum_{a,b}\left(\sum_{i,j}\lambda^a_{ji}\lambda^b_{ij}\right)\left(\sum_{k,l}\lambda^a_{lk}\lambda^b_{kl}\right) \tag{6}$$

Recognizing the Trace

Each parenthetical factor is a matrix product summed over both indices — that's a trace!

$$\sum_{i,j}\lambda^a_{ji}\lambda^b_{ij} = \sum_j\left(\sum_i\lambda^b_{ij}\lambda^a_{ji}\right) = \sum_j(\lambda^b\lambda^a)_{jj} = \mathrm{Tr}(\lambda^b\lambda^a) \tag{7}$$

Using the standard trace orthogonality of Gell-Mann matrices:

$$\mathrm{Tr}(\lambda^a\lambda^b) = 2\,\delta^{ab} \tag{8}$$

we get:

$$\langle|C|^2\rangle = \frac{1}{144}\sum_{a,b}(2\delta^{ab})(2\delta^{ab}) = \frac{1}{144}\sum_{a,b}4\,\delta^{ab}\delta^{ab}$$

$$= \frac{4}{144}\sum_{a=1}^{8}1 = \frac{4\times 8}{144} = \frac{32}{144} \tag{9}$$

Color-Averaged Color Factor

$$\boxed{\langle |C|^2 \rangle = \frac{2}{9}} \tag{10}$$

This factor $\frac{2}{9}$ is universal for all four processes qq, $q\bar{q}$, $\bar{q}q$, $\bar{q}\bar{q} \to$ same, via single gluon exchange (Thomson eq. 10.29; Griffiths Section 9.2).

4.2 Option B: Specific Color Choices

Alternatively, we can pick specific colors. For example, suppose:

- Incoming: q is red ($i = 1$), q' is green ($k = 2$)
- Outgoing: same colors ($j = 1$, $l = 2$), since this is the elastic channel

Then the color factor at amplitude level is:

$$C = \frac{1}{4} \sum_{a=1}^{8} \lambda^a_{11} \lambda^a_{22} \tag{11}$$

Only λ^3 and λ^8 have non-zero entries in both the 11 and 22 positions:

$$C = \frac{1}{4} \left[\lambda^3_{11}\lambda^3_{22} + \lambda^8_{11}\lambda^8_{22} \right] = \frac{1}{4} \left[(1)(-1) + \left(\frac{1}{\sqrt{3}}\right)\left(\frac{1}{\sqrt{3}}\right) \right] = \frac{1}{4} \left[-1 + \frac{1}{3} \right] = -\frac{1}{6} \tag{12}$$

Physical Meaning of the Sign

The negative color factor for $rg \to rg$ means the amplitude has the opposite sign compared to the QED case. Whether this is attractive or repulsive depends on the full context (for qq scattering via t-channel exchange, a negative color factor corresponds to an *attractive* interaction in certain color channels).

For this specific color choice, $|C|^2 = 1/36$. Other elastic color choices give different individual values (e.g., $rr \to rr$ gives $C = +1/3$, so $|C|^2 = 1/9$), but the *average* over all initial colors and sum over all final colors always gives $2/9$.

5 Step 4: Assemble the Final Result

Putting together the coupling constant replacement and the color factor with the QED spinor trace result:

$$\langle |\mathcal{M}|^2 \rangle_{\text{QCD}} = \underbrace{\frac{g_s^4}{16}}_{\substack{\text{from vertex} \\ \text{factors}}} \times \underbrace{\frac{2}{9}}_{\substack{\text{color} \\ \text{factor}}} \times \underbrace{\frac{1}{g_e^4} \cdot 2g_e^4 \frac{s^2+u^2}{t^2}}_{\substack{\text{QED result with} \\ g_e^4 \text{ stripped}}} \tag{13}$$

Wait — let's be more careful. The QED result is:

$$\langle |\mathcal{M}|^2 \rangle_{\text{QED}} = 2g_e^4 \frac{s^2+u^2}{t^2}$$

where g_e^2 came from two vertex factors each contributing g_e, i.e., $g_e^4 = (g_e)^4$.

In QCD, the vertex factor $g_e\gamma^\mu$ is replaced by $\frac{g_s}{2}\lambda^a\gamma^\mu$. So everywhere g_e appeared, we replace $g_e \to \frac{g_s}{2}$ (for the coupling part) and multiply by the color factor. This means:

$$g_e^4 \to \left(\frac{g_s}{2}\right)^4 \times (\text{color factor from } \lambda\text{'s})$$

The factor of $\frac{1}{4}$ from spin averaging is the same (quarks are spin-$\frac{1}{2}$), and additionally we average over $3 \times 3 = 9$ initial color states.

Combining everything:

Final Result: $\langle|\mathcal{M}|^2\rangle$ for $q\,q' \to q\,q'$

$$\boxed{\langle|\mathcal{M}|^2\rangle_{qq'\to qq'} = \frac{4}{9}\,g_s^4\,\frac{s^2+u^2}{t^2}} \tag{14}$$

Derivation of the prefactor:

$$\frac{g_s^4}{16} \times \frac{2}{9} \times 2 \times 16 = \frac{4}{9}\,g_s^4$$

where the factor of 16 in the numerator comes from the QED trace calculation (the factor multiplying g_e^4 in the traces gives $2 \times 16 = 32$, divided by 16 from the vertex couplings).

Equivalently, in terms of $\alpha_s = g_s^2/4\pi$:

$$\langle|\mathcal{M}|^2\rangle = \frac{4}{9}\,(4\pi\alpha_s)^2\,\frac{s^2+u^2}{t^2} \tag{15}$$

Cleanest Way to See It

The simplest derivation is by direct scaling. Define the "stripped" QED result (no coupling) as the spin-trace factor $T = 2\frac{s^2+u^2}{t^2}$. Then:

$$\text{QED:} \quad \langle|\mathcal{M}|^2\rangle = g_e^4 \cdot T \tag{16}$$

$$\text{QCD:} \quad \langle|\mathcal{M}|^2\rangle = \left(\frac{g_s}{2}\right)^4 \cdot \langle|C|^2\rangle \cdot T = \frac{g_s^4}{16} \cdot \frac{2}{9} \cdot T \tag{17}$$

But wait, we also need the $\frac{1}{9}$ color averaging, which is already included in $\langle|C|^2\rangle = 2/9$.
So: $\frac{g_s^4}{16} \times \frac{2}{9} \times T = \frac{g_s^4}{72} \times 2\frac{s^2+u^2}{t^2}$...
Actually, let us be completely explicit. The QED amplitude squared (before spin averaging) contains g_e^4/t^2 from $(g_e^2/t)^2$. After Casimir's trick (sum over all spins), the trace calculation gives a factor. Then $\frac{1}{4}$ for spin averaging. Let's track factors from scratch.

6 Step 5: Careful Factor Tracking from Scratch

QED Amplitude (Griffiths 7.104)

$$\mathcal{M}_{\text{QED}} = -\frac{g_e^2}{t}\,[\bar{u}(3)\gamma^\mu u(1)][\bar{u}(4)\gamma_\mu u(2)]$$

Squaring and spin-summing via Casimir's trick:

$$\sum_{\text{spins}} |\mathcal{M}|^2 = \frac{g_e^4}{t^2}\,\text{Tr}[\gamma^\mu \not{p}_1 \gamma^\nu \not{p}_3]\,\text{Tr}[\gamma_\mu \not{p}_2 \gamma_\nu \not{p}_4]$$

The trace calculation (massless limit) gives:

$$\sum_{\text{spins}} |\mathcal{M}|^2 = \frac{g_e^4}{t^2} \cdot 32\big[(p_1 \cdot p_2)(p_3 \cdot p_4) + (p_1 \cdot p_4)(p_2 \cdot p_3)\big]$$

In Mandelstam variables (massless): $p_1 \cdot p_2 = s/2$, $p_1 \cdot p_4 = u/2$, etc., so:

$$\sum_{\text{spins}} |\mathcal{M}|^2 = \frac{g_e^4}{t^2} \cdot 32 \cdot \frac{s^2+u^2}{4} = \frac{8g_e^4(s^2+u^2)}{t^2}$$

Spin averaging ($\frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$ for two spin-$\frac{1}{2}$ particles):

$$\langle|\mathcal{M}|^2\rangle_{\text{QED}} = \frac{1}{4} \cdot \frac{8g_e^4(s^2+u^2)}{t^2} = \frac{2g_e^4(s^2+u^2)}{t^2} \quad \checkmark$$

QCD Amplitude (Griffiths 9.31)

$$\mathcal{M}_{\text{QCD}} = -\frac{g_s^2}{4}\frac{1}{t}\,[\bar{u}(3)\gamma^\mu u(1)][\bar{u}(4)\gamma_\mu u(2)]\;\sum_a \lambda^a_{ji}\lambda^a_{lk}$$

Squaring and spin-summing (Dirac traces identical to QED):

$$\sum_{\text{spins}} |\mathcal{M}|^2 = \frac{g_s^4}{16}\frac{1}{t^2}\cdot 32\cdot\frac{s^2+u^2}{4}\;\times\;\left|\sum_a \lambda^a_{ji}\lambda^a_{lk}\right|^2$$

$$= \frac{g_s^4}{16}\cdot\frac{8(s^2+u^2)}{t^2}\times\sum_{a,b}\lambda^a_{ji}\lambda^a_{lk}\lambda^b_{ij}\lambda^b_{kl}$$

Now average over spins ($\times\frac{1}{4}$) AND average over initial colors ($\times\frac{1}{9}$), sum over final colors:

$$\langle|\mathcal{M}|^2\rangle_{\text{QCD}} = \frac{1}{4}\cdot\frac{1}{9}\cdot\frac{g_s^4}{16}\cdot\frac{8(s^2+u^2)}{t^2}\;\sum_{a,b}\sum_{i,j,k,l}\lambda^a_{ji}\lambda^b_{ij}\lambda^a_{lk}\lambda^b_{kl}$$

$$= \frac{1}{4}\cdot\frac{1}{9}\cdot\frac{g_s^4}{16}\cdot\frac{8(s^2+u^2)}{t^2}\;\sum_{a,b}\text{Tr}(\lambda^a\lambda^b)\,\text{Tr}(\lambda^a\lambda^b)$$

$$= \frac{1}{4}\cdot\frac{1}{9}\cdot\frac{g_s^4}{16}\cdot\frac{8(s^2+u^2)}{t^2}\;\sum_{a,b}(2\delta^{ab})^2$$

$$= \frac{1}{4}\cdot\frac{1}{9}\cdot\frac{g_s^4}{16}\cdot\frac{8(s^2+u^2)}{t^2}\cdot 4\cdot 8$$

$$= \frac{1}{4}\cdot\frac{1}{9}\cdot\frac{g_s^4}{16}\cdot\frac{8(s^2+u^2)}{t^2}\cdot 32 \tag{18}$$

Computing the numerical prefactor:

$$\frac{1}{4}\cdot\frac{1}{9}\cdot\frac{1}{16}\cdot 8\cdot 32 = \frac{256}{576} = \frac{4}{9}$$

Final Answer

$$\boxed{\langle|\mathcal{M}|^2\rangle_{qq'\to qq'} = \frac{4}{9}\,g_s^4\,\frac{s^2+u^2}{t^2}} \tag{19}$$

This is the standard textbook result for different-flavor quark–quark scattering via single gluon exchange (cf. Thomson eq. 10.29, Halzen & Martin Section 10.4).

Comparison with QED: The ratio is

$$\frac{\langle|\mathcal{M}|^2\rangle_{\text{QCD}}}{\langle|\mathcal{M}|^2\rangle_{\text{QED}}} = \frac{4}{9}\cdot\frac{g_s^4}{2g_e^4} = \frac{2}{9}\cdot\frac{g_s^4}{g_e^4}$$

The factor $\frac{2}{9}$ is precisely the color-averaged color factor, and the g_s^4/g_e^4 is the coupling constant replacement (with the $\frac{1}{2}$ from each QCD vertex already absorbed).

7 Summary of Key Ideas

What You Should Take Away

1. **QCD inherits QED's Dirac structure.** The spinor traces (Casimir's trick, contraction theorems) are identical. The s^2+u^2 in the numerator comes entirely from the spin/momentum structure, which is the same for any spin-$\frac{1}{2}$ particle exchanging a massless spin-1 boson.

2. **Color is a multiplicative overlay.** The color factor is computed independently of the spin traces, then multiplied in. This is because λ^a_{ji} are just numbers (elements of 3×3 matrices), not 4×4 Dirac matrices.

3. **The trace identity $\mathbf{Tr}(\lambda^a\lambda^b) = 2\delta^{ab}$** is the workhorse of color factor calculations. It's the SU(3) analog of $\mathrm{Tr}(\sigma^i\sigma^j) = 2\delta^{ij}$ for Pauli matrices.

4. **$\frac{2}{9}$ is universal** for single-gluon-exchange between any pair of quarks/antiquarks, when color-averaged.

5. **Different flavors $\Rightarrow$ t-channel only.** Same flavors would require a u-channel diagram (antisymmetrized), which is why the problem specifies different flavors.